Dual-Task ConvLSTM-UNet for Instance Segmentation of Weakly Annotated Microscopy Videos

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***Abstract*— Convolutional Neural Networks (CNNs) are considered state of the art segmentation methods for biomedical images in general and microscopy sequences of living cells, in particular. The success of the CNNs is attributed to their ability to capture the structural properties of the data, which enables accommodating complex spatial structures of the cells, low contrast, and unclear boundaries. However, in their standard form CNNs do not exploit the temporal information available in time-lapse sequences, which can be crucial to separating touching and partially overlapping cell instances. In this work, we exploit cell dynamics using a novel CNN architecture which allows multi-scale spatio-temporal feature extraction. Specifically, a novel recurrent neural network (RNN) architecture is proposed based on the integration of a Convolutional Long Short Term Memory (ConvLSTM) network with the U-Net. The proposed ConvLSTM-UNet network is constructed as a dual-task network to enable training with weakly annotated data, in the form of approximate cell centers, termed markers, when the complete cells' outlines are not available. We further use the fast marching method to facilitate the partitioning of clustered cells into individual connected components. Finally, we suggest an adaptation of the method for 3D microscopy sequences without drastically increasing the computational load. The method was evaluated on the Cell Segmentation Benchmark and was ranked among the top three methods on six submitted datasets. Exploiting the proposed built-in marker estimator we also present state-of-the-art cell detection results for an additional, publicly available, weekly annotated dataset. The source code is available at `https://gitlab.com/shaked0/lstmUnet`.**

***Index Terms*— Deep learning, image sequences, image segmentation, recurrent neural networks, microscopy, neural networks, object segmentation.**

I. Introduction

LIVE cell imaging covers a wide range of techniques enabling the quantitative analysis of cellular processes. Recent studies in systems biology concentrate on single-cell analysis, which requires calculating properties such as cell size, shape, mobility, and protein level for each individual cell. Extracting these properties requires the separate segmentation of each cell. This is referred to as *instance* segmentation, which differs from semantic segmentation, where all cells are combined into a single group (foreground-background). Instance segmentation is considered a challenging task since the sequences are made up of multiple (sometimes hundreds or even thousands) visually similar and possibly overlapping objects. The process of annotating each cell instance is, therefore, error-prone and exhaustive. Model-based approaches can address the difficulties inherent to live-cell imaging analysis [1]–[11]. However, due to the high variability in datasets, experimental protocols, and imaging techniques (fluorescent, phase contrast, differential interference contrast, etc.), reliance on specific characteristics of the data becomes a weakness.

Data-driven approaches, on the other hand, and Deep Learning (DL) specifically can address a huge variety of datasets by implicitly constructing a model from the data [12]–[16], [16]–[29]. Convolutional Neural Networks (CNNs) are considered state-of-the-art image segmentation methods as they can learn the spatial properties of the objects of interest through training. The most common CNN architecture for microscopy cell segmentation is the U-Net [14], an encoder-decoder based architecture with skip connections. The U-Net captures features at multiple scales and propagates them from each encoder layer to the corresponding decoder layer. One property of CNNs, the U-Net included, is their spatial invariance, which means that the spatial location is not a defining property. However, when considering *instance* segmentation, this invariance limits the network ability to distinguish between visually similar objects in different locations. The main difficulty is related to separating adjacent cells. One approach is to consider the instance segmentation problem as semantic segmentation with additional constraints. In the classical U-Net paper [14], background pixels were weighted inversely proportional to their proximity to the nearest cells to constrain them to separate cell instances. In a more recent paper by [30], a new loss formulation, which facilitates the separation of touching cells and supports class imbalance, was proposed. The key idea is the incorporation of Youden's J statistic regularization term into the loss function.

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The majority of existing machine learning approaches for cell segmentation are supervised and rely on accurate instance-level annotations throughout the sequence. These annotations are usually provided by experts and are considered as gold-truth (GT) segmentation. In the current work, we adopt the approach we presented in [12] and formulate an instance segmentation that partitions the image into three distinct *semantic* regions; namely the background, foreground, and cell boundary, where the latter separates foreground instances. Obviously, dense expert segmentation is not feasible, and most of the training datasets include sparse, occasional full cell outlines where many of the cells are only annotated by pointwise markers, approximately at their centers. We refer to this type of weak annotations as 'markers,' which are much easier to obtain, faster to annotate, and still informative [19].

The cell boundary class facilitates partitioning the entire cell region into individual cell instances; however, it may not be sufficient when the cells touch, partially overlap, or vanish. Several classical methods demonstrated the advantage of utilizing the time domain to predict cell appearance and boundaries throughout the sequence of cell instances [1]–[4].

More recent works exploit cell dynamics to enhance single cell detection capabilities as a part of a deep learning framework. Hayashida *et al.* proposed a cell tracking method called Cell Motion Field (CMF) [31]. Cell detection was performed as a preliminary step to the task of cell-to-cell association. This step eas accomplished by training a U-Net as a regressor to predict cell positions. In a later work of this group, called Motion and Position Map (MPM), cell detection and association were performed simultaneously by predicting cell motions and locations using the U-Net architecture [32]. Very recently, the MPM framework has been extended to include long term motion, thus improving both association and detection [33].

An alternative approach to exploit the temporal information embedded in the sequential data itself is by using a Recurrent Neural Network (RNN). An RNN is equipped with feedback connections and can therefore capture dynamic behavior. The network architecture proposed in [34] incorporates convolutional gated recurrent units (ConvGRU) into a stacked hourglass network to utilize temporal information. In addition, they defined an embedding loss based on cosine similarities, such that the network predicts unique embeddings for every instance throughout the sequence which is invariant to structural dynamic changes such as cell mitosis.

In the proposed framework we use a special configuration of RNN called Long Short Term Memory (LSTM) network. An LSTM includes an internal memory state vector with gating operations, thus stabilizing the training process [35]. Convolutional LSTMs (Conv-LSTMs) accommodate locally spatial information in image sequences by replacing matrix multiplication with convolutions [36]. A key contribution of the proposed framework is the integration of the well established U-Net with Conv-LSTM. The proposed neural network, called ConvLSTM-UNet, can learn spatio-temporal features at multiple scales and propagate them through time.

One approach to tackle a limited amount of data is adversarial networks. In [12], [37] an adversarial loss function was used to facilitate cell segmentation. For example, in [12], a conditional adversarial loss was constructed by providing pairs of input images and their corresponding segmentations. In [38] the images were transformed from the source microscopy modality to a target domain, assuming that the segmentation would be easier or that more annotated data would be available in the target domain. However, these methods do not make any use of partial annotations.

In our preliminary work [39] an instance segmentation of cells was performed, where the network was trained using the complete cell outlines. Here, we extend our methodological contribution and consider weak annotations in the form of cell markers in training. We further utilize cell markers to separate touching cell instances. Specifically, we structure the ConvLSTM-UNet as a dual-task network, designed to provide two outputs: the segmentation and the marker estimation. Constructing a loss function based on markers, in addition to the cross-entropy loss based on the full segmentations, allows the network to learn cell instances that are only partly annotated. Not only does the marker estimation improves the segmentation, but also facilitates the separation of individual instances at test time. This is achieved by using the Fast Marching (FM) method [40] to calculate the shortest paths between the predicted cell markers and the foreground pixels/voxels of the estimated segmentation map.

In addition, we address the issue of 3D+t image sequences and propose a computationally efficient method for applying our framework to volumetric data. Our method relaxes the need for full 3D annotations and simplifies the data collection with regard to 3D datasets.

The paper's contribution is three-fold. We propose the integration of ConvLSTMs into the U-Net, which enables the incorporation of temporal information to facilitate the segmentation. We further adapt the network to accommodate sparse annotations and allow for more feasible data collection. It is, therefore, suitable for cases where cell annotations are not available. Finally, we propose a method to apply the ConvLSTM-UNet to 3D data while keeping within reasonable computational requirements.

Our method is assessed using time-lapse microscopy data, where both cell dynamics and their spatial properties are considered. Specifically, we tested our method on the Cell Segmentation Benchmark (CSB): `http://www.celltrackingchallenge.net`. At the time of submission, our method was ranked in the top three by the benchmark organizers over several submitted datasets, namely, Fluo-C2DL-Huh7, Fluo-N2DL-HeLa,PhC-C2DH-U373, Fluo-C3DH-A549-SIM, Fluo-N2DH-SIM+ and Fluo-N2DH-SIM+. We further assessed our method for the task of cell detection using an additional public dataset which is composed of high-throughput time-lapse sequences of C2C12 mouse myoblast cells acquired by phase-contrast microscopy under four growth factor conditions [41]. GT annotations of the C2C12 sequences include only the cell center coordinates. Our detection results outperform those obtained by other methods that reported state-of-the-art results for this dataset.

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The rest of the paper is organized as follows. Section II presents a probabilistic formulation of the problem and elaborates on the proposed network. Technical aspects are detailed in Section III. In Section IV we present the data used for the validation and assessment of the proposed method. In Section V we conduct an ablation study to understand the contribution of each component of the proposed framework and demonstrate the strength of our method by presenting competitive cell segmentation and detection results. We conclude in Section VI.

II. Methods

The main challenge of individual cell segmentation from microscopy sequences is the separation of tightly clustered cells. We address this multiple instance separation using a twofold approach. First, the segmentation representation is designed to enhance the delineation of individual cells by a partitioning of the image domain into three classes: background, foreground and cell boundaries using a three-class cross-entropy loss function for segmentation [12], [39]. Second, inspired by [19], we aim to approximately detect cell centers (cell markers). The markers, as opposed to the full segmentation, do not cover the entire cell but rather are small "blobs" somewhere within the cells. They have two desirable properties. First, each individual cell instance is exclusively identified by a single marker (and vice versa) since one marker will never overlap or touch boundaries with a neighboring marker. Second, the markers are easier and faster to annotate, enabling the annotator to provide ample training data. In fact, an annotation by markers is much more feasible and, therefore, more common in most laboratories, in particular, for 3D data. We train the network to estimate both the segmentation and the markers and connect the two using the Fast Marching (FM) method [40]. Fig. 1 presents an overview of the proposed framework with a visualization of the intermediate steps.

A. Input and Output

The input to the ConvLSTM-UNet network is a sequence of live cell microscopy images of arbitrary length T, as illustrated in Fig. 1A. We define the d dimensional (2 or 3) image domain by $\Omega \in \mathbb{R}^d$. We denote a frame in the input image sequence as $I_t : \Omega \rightarrow \mathbb{R}$, where the subscript $t \in \{0, T-1\}$ denotes the frame index and $I_t(\mathbf{v})$ is the intensity of a pixel (or voxel), $\mathbf{v} \in \Omega$. The output of the network consists of two components, the scalar marker map (Section II-D) $\mathcal{M}_t : \Omega \rightarrow [0, 1]$ which represents the probability of a pixel (voxel) to belong to a marker (cell segmentation core) and the soft segmentation map (Section II-C) denoted as $\mathcal{S}_t : \Omega \rightarrow \{0, 1\}^3$ which represents the probabilities of each pixel (voxel) to belong to the foreground, background or cell boundary. These two maps are then passed to an instance segmentation block (Section II-E) which outputs the final labeled segmentation map, $\Gamma_t : \Omega \rightarrow \mathbb{N}^+$.

B. LSTM-UNet

The proposed network incorporates ConvLSTM [36] blocks into the U-Net architecture [14]. This combination was first suggested in our preliminary work [39] and was shown to be powerful. The U-Net architecture, composed of an encoder-decoder with skip connections, serves to extract meaningful descriptors at multiple image scales. However, this alone does not take into account the dynamics of individual cells, which can facilitate the segmentation. The introduction of ConvLSTM blocks into the network allows us to consider past cell appearances at multiple scales by storing their compact representations in the ConvLSTM memory units. Here, we propose the incorporation of ConvLSTM layers at every scale of the encoder section of the U-Net. The specific architecture was selected based on our preliminary work, which shows its empirical advantage over other alternatives [39]. The network is fully convolutional. It can, therefore, be used with any image size[1] during both training and testing. Fig. 1B illustrates the network architecture. Further technical details are presented in Section III.

The network consists of two sections of N blocks each: the encoder recurrent block $E_{\theta_n}^{\{n\}}(\cdot)$ and the decoder block $D_{\theta_n}^{\{n\}}(\cdot)$, where θ_n are the network parameters. The input to the ConvLSTM encoder layer $n \in \{1, \ldots, N-1\}$ at time $t \in \{0, \ldots, T-1\}$ is composed of the down-sampled output of the previous layer, the output of the current layer at the previous time step, and the ConvLSTM memory cell. We denote these three inputs as $x_t^{\{n\}}$, $h_{t-1}^{\{n\}}$, $c_{t-1}^{\{n\}}$, respectively. Formally we define:

$$(h_t^{\{n\}}, c_t^{\{n\}}) = E_{\theta_n}^{\{n\}}(x_t^{\{n\}}, h_{t-1}^{\{n\}}, c_{t-1}^{\{n\}}) \tag{1}$$

where,

$$x_t^{\{n\}} = \begin{cases} I_t, & n = 0 \\ MaxPool(h_t^{\{n-1\}}), & 0 < n < N \end{cases} \tag{2}$$

The inputs to the decoder layers $n \in \{N, \ldots, 2N-1\}$ are the up-sampled [2] output of the previous layer and the output of the corresponding layer from the encoder denoted by $y_t^{\{n\}}$ and $h_t^{\{2N-1-n\}}$, respectively. We denote the decoder output as $z_t^{\{n\}}$. Formally,

$$y_t^{\{n\}} = \begin{cases} h_t^{\{n-1\}}, & n = N \\ UpSample(z_t^{\{n-1\}}), & N < n \leq 2N-1 \end{cases} \tag{3}$$

where:

$$z_t^{\{n\}} = D_{\theta_n}(y_t^{\{n\}}, h_t^{\{2N-1-n\}}) \tag{4}$$

MaxPool and UpSample refer to the *max pooling* and *up-sampling* operations, respectively. In [39], the network had a single output for the segmentation map. In contrast, here the last layer of the network is split to produce outputs for the soft segmentation map S_t and the marker map M_t, to be discussed in the following sections. The entire network is a composition of N encoder blocks followed by N decoder blocks. Note that the encoder blocks, $E_{\theta_n}^{\{n\}}$, encode high-level **spatio-temporal** features at multiple scales and the decoder blocks, $D_{\theta_n}^{\{n\}}$, refine that information providing full scale segmentation and marker maps.

[1] To avoid artifacts it is preferable to use image sizes that are multiples of eight due to the three max-pooling layers.

[2] We use bi-linear interpolation

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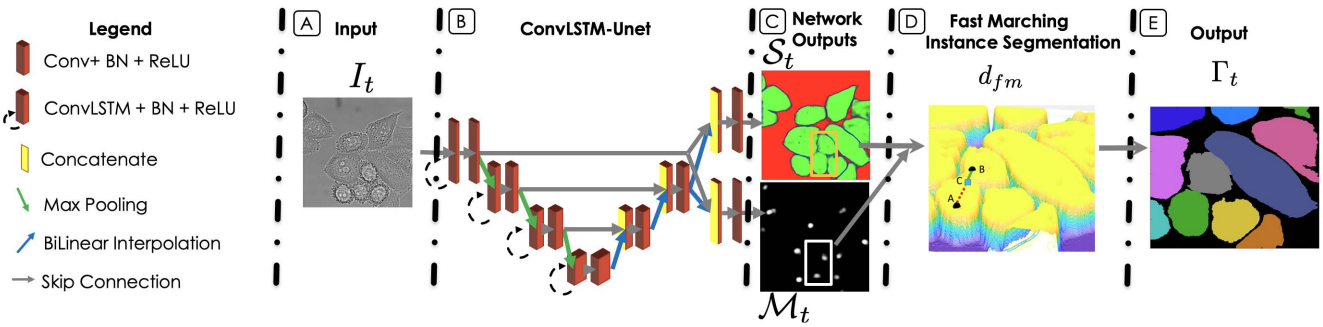

Fig. 1. Method outline A. The input to the ConvLSTM-UNet is a time-lapse sequence of microscopy images. B. The network's down-sampling path consists of ConvLSTM layers of decreasing size, where each is followed by a convolutional layer with ReLU activation. The output is then down-sampled using max-pooling and composes the input to the next layer. The up-sampling path consists of a sequence of layers of increasing size, where the input to each is a concatenation of the lower layer's output and the output of the corresponding layer from the down-sampling path. Each such layer is followed by two convolutional layers with ReLU activation. C. The last layer is split to produce the soft segmentation $\mathcal{S}_t$ and the marker maps $\mathcal{M}_t$. D. The two network outputs are combined using fast marching optimization to refine the instance segmentation. To illustrate the concept of geodesic distance, the cell channel of the U-Net output is presented using a mesh plot. Specifically, the length of geodesic paths between two adjacent cell instances separated by a "ridge" is larger than the length of any path within a single cell instance, regardless of the corresponding Euclidean distances. E. The final label map, Γ_t is constructed by labeling each of the pixels (or voxels) in the segmentation map according to the minimal distance marker.

C. The Soft Segmentation Map

Let θ_S denote the parameters of the soft segmentation layer and let $S_t = D_{\theta_S}(y_t^{\{2N-1\}}, h_t^{\{0\}})$ denote its output. S_t has the same spatial dimension as the input and has three channels corresponding to the non-normalized evidence for the background, foreground and cell boundary classes, encoded in the entries $l \in \{0, 1, 2\}$, respectively. $\mathcal{S}_t$ is constructed based on the softmax equation, as follows:

$$\mathcal{S}_t^l(\mathbf{v}) \triangleq p(l|S_t(\mathbf{v})) = \frac{\exp\{[S_t(\mathbf{v})]_l\}}{\sum_{l' \in \{0,1,2\}} \exp\{[S_t(\mathbf{v})]_{l'}\}} \tag{5}$$

We further denote by $\mathcal{S}_t^{l=1} \triangleq \mathcal{S}_t^{cell}$ the cell channel in the network's output. The upper part of Fig. 1C illustrates the soft segmentation map, which is one of the two outputs of the proposed ConvLSTM-UNet.

D. The Marker Map

For the soft segmentation output, we construct the marker layer $M_t = D_{\theta_M}(y_t^{\{2N-1\}}, h_t^{\{0\}})$ where θ_M are the weights of this layer. The marker map is illustrated in the lower part of Fig. 1D. M_t has the same spatial dimension as the input and has one channel corresponding to the non-normalized evidence for the marker. The marker map, $\mathcal{M}_t$ is defined by the following sigmoid function:

$$\mathcal{M}_t(\mathbf{v}) \triangleq \sigma(M_t(\mathbf{v})) = \frac{1}{1 + \exp^{-M_t(\mathbf{v})}}, \tag{6}$$

An ablation study designed to show the effect of the marker map estimation is presented in Section V-D.

E. Instance Segmentation & Fast Marching Algorithm

The instance segmentation block combines the two outputs of the network, $\mathcal{S}_t$ and $\mathcal{M}_t$, into a single, multi-label map as illustrated in Fig. 1D-E. We define the semantic cell region, i.e., the set of pixels/voxels labeled as foreground, by:

$$\mathcal{R}_t^{cell} = \{\mathbf{v} | \arg\max_{l \in \{0,1,2\}} [\mathcal{S}_t(\mathbf{v})]_l = 1\} \tag{7}$$

The boundary class in $\mathcal{S}_t$ is used to partition the cell region, $\mathcal{R}_t^{cell}$, into connected components, where each component should represent a single cell instance. However, because of possible under-segmentation, the number of connected components may not necessarily correspond to the number of blobs in the marker map, which more faithfully represents the true number of cells. Therefore, we use $\mathcal{M}_t$ to facilitate the separation of merged cell instances. We define a binary marker map $\mathcal{R}_t^{blob}$ by applying a threshold on $\mathcal{M}_t$ as follows:

$$\mathcal{R}_t^{blob} = \{\mathbf{v} | \mathcal{M}_t(\mathbf{v}) \geq 0.5\} \tag{8}$$

$\mathcal{R}_t^{blob}$ is partitioned into $\mathcal{N}_t$ connected components, $\{m_i\}_{i=1}^{\mathcal{N}_t}$.

Refining Instance Segmentation via Fast Marching: We define by Γ_t the final label map of the individual cell instances. Γ_t is constructed by matching connected components m_i in the marker map with each $\mathbf{v} \in \mathcal{R}_t^{cell}$ by exploiting the fast marching method [40]. Specifically, we use the fast marching method to compute the shortest geodesic distances, with respect to a velocity map, between marker pixels/voxels and foreground pixels/voxels. The fast marching method is closely related to Dijkstra's method for computing the shortest path on a graph [42], but has the advantage of convergence to a continuous solution. We use $\mathcal{S}_t^{cell}(\mathbf{v})$ (defined in Eq. 5) as the fast marching velocity map. This map is in fact the cell channel output of the network, which (when normalized) defines the probability of each pixel/voxel to belong to the cells region. Therefore, regions with high probability of being part of the foreground are considered 'high speed' regions whereas cell boundaries and background are considered 'low speed' regions. The corresponding label map Γ_t is constructed by considering the geodesic distances determined by $\mathcal{S}_t^{cell}$ between the markers and every pixel/voxel in the map. A pixel/voxel is then labeled (assigned to a particular cell instance) according to the minimal distance markers.

Fig. 2 illustrates the main idea. Fig. 2(a) presents a color-coded 3-class segmentation of a tight collection of cell instances where some form a single connected component.

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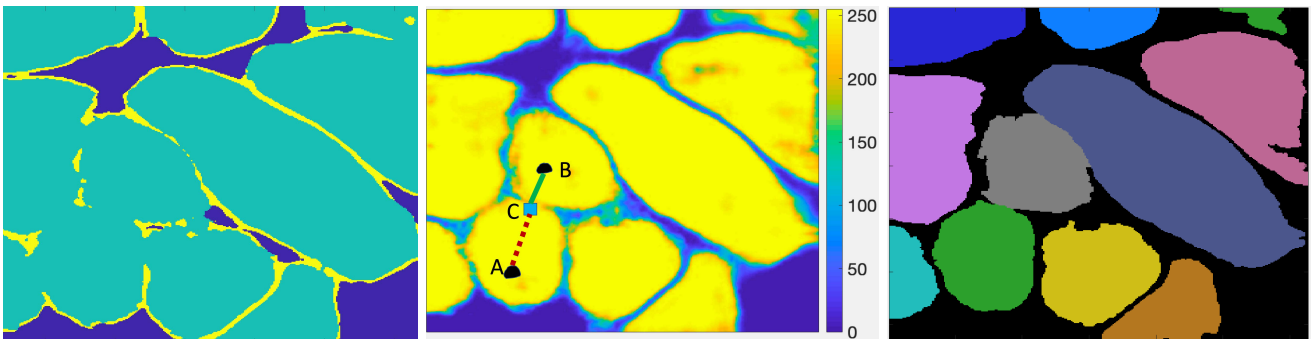

(a) 3-class semantic segmentation (b) Velocity map and cell markers (c) Final label map

Fig. 2. Fast marching for instance segmentation refinement. (a) A color-coded 3-class semantic segmentation of a tight collection of cell instances. The blue, turquoise and yellow indicate the foreground, the cell region and the cell boundaries, respectively, as predicted by the network. (b) A color-coded velocity map of cell instances along with the markers of two cells (*A* and *B*). The velocity map is reconstructed directly from the soft segmentation of the cells ($\mathcal{S}_t^{cell}$) prior to the application of the **max** operator for the generation of the 3-class map shown in (a). The path between some point *C* and marker *A* (dashed red) is shorter (in terms of geodesic distance) than the path from *B* to *C* (green) despite the fact that its Euclidean distance is longer. The shortest paths between the markers and the image pixels/voxels were calculated by fast marching. (c) The label map Γ_t was generated by assigning pixels/voxels to the minimal distance markers. The images shown in this figure are for illustration only. The actual GT markers, provided as part of the training annotations are not presented. Instead, we present only two markers to highlight two specific cells.

The image is partitioned into the background (blue), the cell region (turquoise) and the cell boundaries (yellow). Fig. 2(b) presents a velocity map of the same cell collection along with the markers of two cells (A and B). The velocity map is reconstructed directly from the soft segmentation of the cells ($\mathcal{S}_t^{cell}$) prior to the application of the **max** operator for the generation of the 3-class map shown in Fig. 2(a). Note that the 'length' of the paths (marked in green and red) between each of the markers and some point C is determined by the underlying velocity map (where low speed implies a long path) and not by the Euclidean distance. The label map Γ_t, shown in Fig. 2(c), is generated by assigning pixels/voxels to the minimal distance markers. Formally, the shortest distance between a source point, $\mathbf{v}$, and a target point m_i, given a velocity map $\mathcal{S}_t^{cell}$ in either 2D or 3D, with respect to the input domain is calculated using the following fast marching optimization:

$$\Gamma_t(\mathbf{v}) = \begin{cases} \arg\min\limits_{i=1\ldots\mathcal{N}_t} d_{fm}\left(\mathbf{v}, m_i | \mathcal{S}_t^{cell}\right) & \mathbf{v} \in \mathcal{R}_t^{cell} \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

where d_{fm} denotes a geodesic distance.

F. Addressing the 3D Data

In this paper, we address both 2D and 3D image sequences. Due to the heavy computational load of 3D data, we generate volumetric maps $\mathcal{M}_t$ and $\mathcal{S}_t$ from 2D slices. Let I_t be a 3D frame in an input sequence. I_t is composed of K 2D slices along the depth dimension $\{I_t^{(k)}\}_{k=1}^K$. Similarly, we denote the corresponding soft segmentation and marker maps $\mathcal{S}_t = \{\mathcal{S}_t^{(k)}\}_{k=1}^K$ and $\mathcal{M}_t = \{\mathcal{M}_t^{(k)}\}_{k=1}^K$, respectively. To preserve spatial coherence, we process $I_t^{(k)}$ by feeding the network with three consecutive slices, $\{I_t^{(k-1)}, I_t^{(k)}, I_t^{(k+1)}\}$. These three slices are concatenated along the channel axis, similar to an RGB image. The outputs of the network are the segmentation of the central slice $\mathcal{S}_t^{(k)}$ and its marker map $\mathcal{M}_t^{(k)}$. The two additional slices provide the context in the z axis. The segmentation of the entire volume is constructed by stacking the respective 2D segmentations. We conducted an ablation study to assess this approach for handling 3D data, see Section V-E.

G. Weakly Annotated Data

Manual segmentation of microscopy sequences is a tedious and time-consuming task, particularly for 3D+t data. However, it can be simplified if the annotator marks the approximate cell centers instead. Although one-hot encoded (background, foreground and boundary) labels, $\bar{\mathcal{S}}_t : \Omega \rightarrow \{0, 1\}^3$, for the entire training sequence are preferable, in practice most of the cells in a sequence are only weakly annotated by markers, $\bar{\mathcal{M}}_t : \Omega \rightarrow \{0, 1\}$. Specifically, we assume that at least one frame (or a 2D slice in a 3D volume) is fully annotated. Computing the loss for weakly annotated frames must be handled carefully since 'true' cell regions may not be annotated as such, due to the shortage of full labeling. To address such scenarios, pixels/voxles labeled as background in training frames that contain markers with no corresponding cell masks are marked with flags. For these pixels/voxels the weighted cross entropy loss (Eq. 11) is not considered. Thus the absence of cell labels in the training data does not discourage prediction of cell regions by the network.

H. Training and Loss

The network is trained using Truncated Back Propagation Through Time (TBPTT) [43]. At each backpropagation step, the network is unrolled τ time steps. The total loss for training is the sum of two loss functions: the first for the marker map, $L_{\mathcal{M}}$, and the second for the soft segmentation map, $L_{\mathcal{S}}$. For the marker map, we use the binary cross entropy loss:

$$L_{\mathcal{M}} = -\sum_{t'=t}^{t+\tau} \sum_{\mathbf{v}\in\Omega} [\bar{\mathcal{M}}_{t'}(\mathbf{v}) \cdot \log(\mathcal{M}_{t'}(\mathbf{v})) + (1 - \bar{\mathcal{M}}_{t'}(\mathbf{v})) \cdot \log(1 - \mathcal{M}_{t'}(\mathbf{v}))], \tag{10}$$

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where $\bar{\mathcal{M}}$ is the predicted marker map and $\mathcal{M}$ is the GT. The soft segmentation map is penalized by the weighted, multi-class cross entropy loss, giving higher weights, w_l to the cell contour pixels, due to the class imbalance:

$$L_{\mathcal{S}} = -\sum_{t'=t}^{t+\tau} \sum_{\mathbf{v}\in\Lambda} \sum_{l\in\{0,1,2\}} w_l \cdot [\bar{\mathcal{S}}_{t'}(\mathbf{v})]_l \cdot \log([\mathcal{S}_{t'}(\mathbf{v})]_l) \quad (11)$$

where $\Lambda \subseteq \Omega$ denote the subset of pixels (voxels) within the frame for which annotations exist. Therefore, frames, or pixels (voxels) within frames, which are not annotated, are not taken into account in the loss calculation. The final loss is set to be the sum of the two:

$$L = L_{\mathcal{M}} + L_{\mathcal{S}} \quad (12)$$

III. Implementation Details

A. Libraries

The method was implemented in Python using Tensorflow for the neural networks section and the scikit-fmm Python library for the FM method implementation.

B. Architecture

The network is based on the U-Net architecture, with four encoder and four decoder blocks. Each of the encoder blocks is composed of a ConvLSTM layer, a leaky ReLU, a convolutional layer, a batch normalization [44], a leaky ReLU, and finally, down-sampled using a MaxPool operation. Each of the decoder blocks consists of a bi-linear interpolation, a concatenation with the corresponding encoder block followed by two sets of convolutional layers, batch normalization [44], and a leaky ReLU. The same architecture is used for both the 2D and 3D datasets. All the ConvLSTM kernels are of size 5×5 and all the convolutional layers use a kernel of size 3×3. The feature depths in the encoder and decoder paths are set to be (128, 256, 512, 1024) and (1024, 512, 256, 128), respectively. All the MaxPool layers use a kernel size of 2×2 without overlap. The last convolutional layer uses a kernel size 1×1 with a depth of 3 for the soft segmentation layer S_t and a depth of 1 for the marker layer, M_t, followed by a softmax or a sigmoid layer, respectively, to produce the final probabilities (see Fig. 1).

C. Training Data

The fully annotated training data provided by the CSB organizers include labeled marker maps and labeled cell masks. As part of the proposed training process, the labeled marker maps are converted to binary maps and the labeled training masks are converted into trinary maps, where background, cell regions and boundary pixels/voxels are labeled as described in Section II-C.

D. Training Regime

We trained the networks on each dataset separately for $200K$ iterations ($300K$ for the 3D datasets) using the ADAM optimizer [45] with a learning rate of 0.0001. The unroll length parameter was set to $\tau = 4$, and the batch size was set to five sequences. The loss function weights w_l (Eq. 11) were set to be 0.15, 0.25 and 0.6 for the background, foreground and cell boundaries, respectively (Section II-H).

IV. Data and Augmentation

A. Cell Segmentation Benchmark Datasets

In the following experiments, we focus on the Cell Segmentation Benchmark (CSB), which is an extensive benchmark for time-lapse live-cell microscopy sequences. The benchmark consists of diverse datasets,[3] both 2D and 3D, most of which are real microscopy image sequences with the exception of three simulated datasets.

Each dataset contains four sequences of varying length, where two are allocated for training and two for the test. Annotation of the sequences was performed by several experts. Since the manual annotations of the test sets are not publicly available, the scores presented for these sets are as published by the challenge organizers and are available at `http://celltrackingchallenge.net/latest-csb-results/`. The GT marker map was constructed such that each cell in the sequence is marked by a set of pixels approximately at its center. We note that complete segmentation of the cells was sparse, and not all of the frames and cells were fully segmented. We converted the cell label maps into a three-label, one-hot encoding, representing the background, foreground, and cell boundaries.

1) Manual Annotations: For the 2D datasets, manual segmentation was performed randomly, such that at least 100 cells and at least two frames in a sequence were fully segmented. After reaching the given limit, the annotators inspected the rest of the frames in the given random order and additionally segmented all difficult, somehow interesting, or previously unseen cell instances. The 3D datasets were manually annotated with similar limitations, but for random slices within the 3D frames. Consequently, there is no complete manual 3D segmentation for any of the cells, except for the three simulated datasets (Fluo-N2DH-SIM+, Fluo-N3DH-SIM+, and Fluo-C3DH-A549-SIM). We refer to the actual cell segmentation (in the case of simulated data) as ground-truth segmentation while manual segmentation is referred to as gold-truth segmentation.

2) Silver Truth Annotations: Since complete manual annotation of cell outlines is not feasible the CSB orgenaziers provided a silver-standard corpus (denoted as silver truth) containing computer-generated segmentation masks. The currently available (Dec. 2021) silver truth annotations were generated by averaging (using a modified version of the label fusion approach) the segmentation results of up to 16 top-performing algorithms that simultaneously achieved the SEG and DET scores over the halves of the reported human performance for both the training and test sequences were selected per dataset.

B. Simulated Data

For some of the ablation study experiments, a considerable amount of sequential, fully annotated training and test data

[3]To assess our framework we used most (17) of the datasets that were included in the challenge data at the time we conducted our experiments.

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was required. In addition, we preferred demonstrating our framework with the actual GT segmentation of the cells. We therefore used the CytoPacq web service[4] [46] to simulate 92 cell sequences 51 of which are 2D (44/7 training/test) and 41 (34/7 training/test) are 3D. Each sequence contains up to 200 frames. The CytoPacq system was tuned to produce data similar to the Fluo-N2DH-SIM+ and Fluo-N3DH-SIM+ datasets from the CSB. We refer to these datasets, using the challenge naming convention as Fluo-N2DH-BGUSIM and Fluo-N3DH-BGUSIM for the 2D and 3D datasets, respectively. These datasets can be downloaded at `https://ieee-dataport.org/documents/bgusim`.

C. C2C12 Myoblast Datasets

The C2C12 dataset [41] is composed of high-throughput time-lapse sequences of C2C12 mouse myoblast cells acquired by phase-contrast microscopy under four growth factor conditions, FGF2, BMP2, FGF2 + BMP2, and control (no growth factor). For each condition, four image sequences were captured, each consists of 780 frames (12 frames-per-second, 1392 × 1040 pixels). The GT annotation includes a list of coordinates of the cell instances centers along with the corresponding indices of the depicted biological cells. There is a complete cell annotation for only one sequence, where only three cells and their lineage trees were marked in the others. For the training we used a total of 400 annotated frames from four different sequences as described in [32]. Fig. 6 presents some examples of the test images along with the GT cell coordinates and marker predictions (by the proposed method).

D. Data Augmentation

The data were randomly augmented both spatially and temporally by:

1) Random horizontal and vertical flip
2) Random 90° rotation,
3) Random crop of size 256 × 256,
4) Random sequence reverse ($\{T, T-1, \ldots, 2, 1\}$),
5) Random temporal down-sampling by a factor of $k \in \{1, 2, 3\}$,
6) Random affine transformations as suggested in [14]. The affine transformation has four degrees of freedom (DOF) including rotations, anisotropic scaling and skew where translation is already obtained by random crops. The affine matrix is calculated based on two randomly selected sets of corresponding points (x, y) and $(x + \Delta_x, y + \Delta_y)$, where $\Delta_x, \Delta_y \in [-20, 20]$.
7) Elastic transformations as suggested in [47]. The displacement fields $\triangle(x, y) = (r_x\alpha, r_y\alpha)$, are generated with randomly selected r_x, r_y in the range of $[-1, 1]$ and α is set to 512 (twice the image size). The displacement fields are spatially smoothed by a convolution with a Gaussian kernel with a standard deviation proportional to the image size.
8) Random contrast with a contrast factor in [0.5, 1.5] and random brightness changes of up to 10% of the max value.
9) For the 3D dataset, a random slice was selected at the beginning of each sequence, and the input frame was padded by one slice before and after the selected slice.

[4] `https://cbia.fi.muni.cz/simulator`

V. Experiments and Results

A. Evaluation Method

The measures for evaluating the Cell Segmentation Benchmark results were defined by the benchmark orgenizers. Specifically, we used the SEG measure for segmentation [48] and the DET measure for detection [49]. The SEG measure is defined as the mean Jaccard index $\frac{|A \cap B|}{|A \cup B|}$ of a pair of GT labeled A and its corresponding, matched, segmentation B. A segmentation is considered a match if $|A \cap B| > \frac{1}{2}|A|$.

The DET measure is based on a comparison of the nodes of acyclic oriented graphs representing cells in both the GT and the evaluated method. This measure is a special case of the normalized Acyclic Oriented Graph Matching (AOGM-D) measure for detection and is fully formulated and described in [49] as follows:

$$\mathrm{D}ET = 1 - \min(\text{AOGM-D}, \text{AOGM-D}_0)/\text{AOGM-D}_0,$$

where AOGM-D is the cost of transforming a set of nodes provided by the participant into a set of GT nodes, and AOGM-D_0 is the cost of creating the set of GT nodes from scratch. The normalization ensures that DET always falls within the interval [0,1].

The OP_{CSB} is the average of the DET and SEG measures, $OP_{CSB} = 0.5(\mathrm{D}ET + \mathrm{S}EG)$.

For evaluating our point level detection results for the C2C12 dataset we used precision $Pr = TP/(TP + FP)$, recall $Rc = TP/P$ and F1-score $F_1 = 2PrRc/(Pr + Rc)$ as were defined in [32]. Specifically, we compared the predicted markers with the GT cell centers, where the true positives (TP) were all cases in which a predicted cell marker includes the GT coordinates of a single cell. A predicted cell marker which did not include cell coordinates was considered as a false positive example (FP) while the total number of cell centers defined the number all positive cases (P).

B. Ablation Studies: Training and Test Datasets

The following ablation study experiments were performed using the simulated Fluo-N2DH-BGUSIM and Fluo-N3DH-BGUSIM datasets, detailed in Section IV-B and two real CSB datasets, namely Fluo-C2DL-Huh7 and DIC-C2DH-HeLa. The Fluo-N2DH-BGUSIM dataset was split into 44 distinct, fully annotated training (44) and test (7) sequences. The Fluo-N3DH-BGUSIM dataset was split into distinct, fully annotated training (34) and test (7) sequences. Since test annotations for the CSB datasets are not available, we used the two training sequences provided for each dataset instead. Specifically, we conducted each experiment twice, training the network using one training sequence and testing on the other (we refer to this as a per-sequence test). The scores of the ablation study experiments for these datasets are the averages of the two runs.

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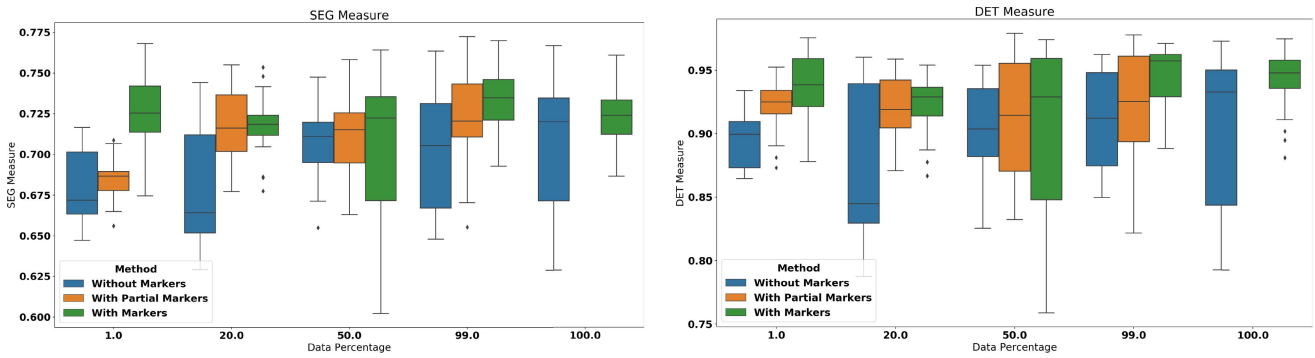

Fig. 3. Box plots showing the SEG and DET measures for the three experimental settings described in Section V-C. It is clear that the additional information from the marker annotations outperforms the baseline across all percentages of annotations. Furthermore, adding the weak annotations for all images of the sequences reaches a high score even for a low percentage of segmentation annotations.

C. Ablation Study: Training With Weak Annotations

Reducing the annotation load is an important goal. We empirically evaluated the significance of the weak annotation proposed here. We refer to two types of annotations, full segmentation and marker annotations. The former is time-consuming and tedious, while the latter is relatively easy to achieve. We examined three different experimental setting of the method. First, as proposed in our preliminary work [39], we tested the ConvLSTM-UNet without the additional marker estimation. We then tested the proposed dual-task network, which further incorporated marker estimation as a complementary output to the segmentation. We trained the ConvLSTM-UNet on the Fluo-N2DH-BGUSIM dataset to evaluate the three different experimental settings of the method. We conducted five sets of experiments, each with a different percentage of annotated training sequences $(1\%, 20\%, 50\%, 99\%, 100\%)$. Fig. 3 shows box plots of the SEG and DET measures for each percentage point, where each box presents segmentation results obtained for three, randomly initialized, training sessions. The blue boxes refer to sets of experiments for which $X\%$ of the training sequences were used and markers were not exploited. For these experiments we used the ConvLSTM-UNet as introduced in [39]. The orange boxes refer to sets of experiments for which $X\%$ of the training sequences were used including both segmentation masks and markers. The green boxes refer to sets of experiments for which all training sequences were used where $X\%$ of the training annotations included both markers and segmentations and $(100-X)\%$ of the training annotations included only markers. Note that the orange boxes (their averages) are higher than the blue boxes which implies that training with both markers and cell outlines (segmentations) is preferable over using segmentations alone. Note also that the green boxes (their averages) are higher than the orange boxes which implies that using cells annotated solely with markers in addition to the fully annotated training data improves the training and therefore the segmentation results. While the performances of the ConvLSTM-UNet network, without the weak annotations, degraded as we reduced the percentage of full segmentations, the networks that utilized the weak annotations managed to compensate for this shortage. These findings demonstrate the contribution of weak annotations, suggesting that much of the manual annotation load can be reduced.

D. Ablation Study: Instance Segmentation Refinement

Next, we examined the contribution of the marker estimation and the fast marching algorithm to the instance segmentation accuracy. Again, we compared three different experimental settings. First, we tested the ConvLSTM-UNet without marker estimation (as proposed in our previous work [39]). We then trained the network using the marker estimation. However, in the second experimental setting we did not use the marker outputs at test time and relied solely on the segmentation channel. In the third experiment we used the markers for both enhancing the training and separating adjacent cell instances (via fast marching) during the inference, as presented in Section II-D. Table I presents the average SEG and DET scores obtained for the Fluo-C2DL-Huh7 and DIC-C2DH-HeLa and those obtained for the Fluo-N2DH-BGUSIM test dataset (average score obtained for three runs). The results demonstrate the advantage of the marker estimation over training without the markers. The incorporation of the fast marching method further enhanced the model, if, as in the case of the Fluo-C2DL-Huh7 and DIC-C2DH-HeLa, some of the cells are clustered together. Visual assessment is provided in Fig. 4 for a frame from the Fluo-C2DL-Huh7 and DIC-C2DH-HeLa **per-sequence test** evaluation.

E. Ablation Study: 3D Slice Input Padding

We next examined the influence of the number of input slices for the segmentation of 3D data. For this set of experiments we use the Fluo-N3DH-BGUSIM dataset that was split into distinct, fully annotated training(34) and test(7) sequences. We trained the networks **without** the marker map and evaluated the segmentation alone. We tested two configurations using either single input slices or three-slice configuration, as presented in Section II-F. The computational load of the method can be approximated by the number of weights in the network because each additional weight requires two additional operations, namely, multiplication and addition. The results presented in Table II clearly show the

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TABLE I

MARKER ESTIMATION EXPERIMENTS EXPERIMENTAL RESULTS OBTAINED FOR THE FLUO-C2DL-HUH7, DIC-C2DH-HELA AND FLUO-N2DH-BGUSIM DATASETS FOR THE EVALUATION OF THE PROPOSED DUAL-TASK CONVLSTM-UNET FRAMEWORK. EACH ROW IN THE TABLE PRESENTS SEG AND DET SCORES FOR EACH OF THE THREE EXPERIMENTAL SETTINGS DESCRIBED IN SECTION V-D. THE TOP ROW PRESENTS THE RESULTS OBTAINED USING CONVLSTM-UNET ARCHITECTURE TRAINED WITHOUT MARKER ESTIMATION. THE SECOND ROW PRESENTS THE SEGMENTATION RESULTS PRIOR TO THE APPLICATION OF THE INSTANCE SEGMENTATION BLOCK II-E. THE THIRD ROW DEMONSTRATES THE ADVANTAGE OF USING FAST MARCHING FOR INSTANCE SEGMENTATION REFINEMENT. VISUAL SEGMENTATION RESULTS FOR THE FLUO-C2DL-HUH7 AND DIC-C2DH-HELA ARE SHOWN IN FIG. 4

	Fluo-C2DL-Huh7		DIC-C2DH-HeLa		FLUO-N2DH-BGUSIM	
Configuration	SEG	DET	SEG	DET	SEG	DET
Without Marker Estimation	0.619 (±0.124)	0.854 (±0.065)	0.733 (±0.004)	0.901 (±0.037)	0.745 (±0.013)	0.941 (±0.013)
Trained with Marker Estimation, Tested Without	0.656 (±0.093)	0.871 (±0.050)	0.771 (±0.019)	0.904 (±0.038)	0.758 (±0.015)	**0.952** (±0.013)
Trained and Tested with Marker Estimation	**0.751** (±0.037)	**0.962** (±0.018)	**0.826** (±0.004)	**0.949** (±0.031)	**0.759** (±0.016)	**0.952** (±0.013)

TABLE II

3D SLICE INPUT EXPERIMENTS COMPARISON OF INPUT TYPES FOR THE 3D DATA EVALUATED ON OUR FLUO-N3DH-BGUSIM DATASET. THE 3-SLICE CONFIGURATION CLEARLY PERFORMED BETTER FOR BOTH SEG AND DET MEASURES WITH ONLY A MINOR ADDITION OF 3200 TO THE TOTAL NUMBER OF WEIGHTS, $|\Theta|$

Configuration	SEG	DET	$\lvert\Theta\rvert$
Single Slice	0.525 (±0.027)	0.897 (±0.036)	9,770,211
Three Slices	**0.638** (±0.017)	**0.949** (±0.017)	9,773,411

TABLE III

COMPARISON OF THE TEST SEGMENTATION SCORES OBTAINED WHERE THE PROPOSED CONVLSTM-UNET IS TRAINED WITH DIFFERENT CONFIGURATIONS OF THE CE LOSS WEIGHTS ([BACKGROUND (BG), FOREGROUND (FG), CELL BOUNDARY (CB)]). THE STUDY WAS CONDUCTED FOR THE DIC-C2DH-HELA AND FLUO-C2DL-HUH7 DATASETS USING THE PER-SEQUENCE TESTING METHODOLOGY. REFER TO THE TEXT FOR MORE DETAILS

	DIC-C2DH-HeLa		Fluo-C2DL-Huh7	
Class Weights [BG, FG, CB]	SEG (std)	DET (std)	SEG (std)	DET (std)
[0.15, 0.25, 0.6]	0.826 (±0.004)	0.949 (±0.032)	0.751 (±0.037)	0.962 (±0.018)
[0.33, 0.33, 0.33]	0.829 (±0.002)	0.940 (±0.027)	0.739 (±0.048)	0.955 (±0.024)
[0.1, 0.2, 0.7]	0.825 (±0.008)	0.946 (±0.034)	0.746 (±0.051)	0.957 (±0.024)

advantages of the three-slice input. In summary, while the overall computational complexity of the 3-slice input is only increased by 0.032%, which is still very close to a regular slice-by-slice (2D) network, the gain is significant.

F. Ablation Study: Cross Entropy Loss Weights

By setting different weights for the background, foreground and cell boundaries in the cross entropy loss function (Eq. 11) we aimed to compensate for the class imbalance. The number of boundary pixels/voxels is much smaller than the number of pixels/voxels labeled as cells. In addition, for most datasets the background region is larger than the foreground. Note that although the CBS datasets vary significantly we have not tuned these parameters for the different sets. To demonstrate the effect of the loss function weights, we conducted an ablation study experiment in which we tested three different weight configurations for two different CSB datasets, namely, DIC-N2DL-HeLa and Fluo-C2DL-Huh7. The results presented in Table III shows that for these datasets, neither configuration was advantageous for both the SEG and the DET scores.

TABLE IV

COMPARISON OF THE TEST SEGMENTATION SCORES OBTAINED FOLLOWING TRAINING OF THE PROPOSED CONVLSTM-UNET WITH AND WITHOUT TIME FLIP AUGMENTATIONS. THE STUDY WAS CONDUCTED FOR THE DIC-C2DH-HELA AND FLUO-C2DL-HUH7 DATASETS USING THE PER-SEQUENCE TESTING METHODOLOGY. REFER TO THE TEXT FOR MORE DETAILS

	DIC-C2DH-HeLa		Fluo-C2DL-Huh7	
Configuration	SEG (std)	DET (std)	SEG (std)	DET (std)
With Time Aug	0.826 (±0.004)	0.949 (±0.032)	0.751 (±0.037)	0.962 (±0.018)
w/o Time Aug	0.824 (±0.002)	0.945 (±0.033)	0.729 (±0.023)	0.959 (±0.011)

G. Ablation Study: Temporal Augmentation

We apply a temporal augmentation by reversing the temporal sequences, which allows us to increase and further diversify the training sequences. We conducted an ablation study to demonstrate that while the reverse sequences simulate improbable events of merging cells (rather than splitting) this temporal augmentation is still beneficial. The results presented in Table IV which were obtained with and without time flips show that for sparsely annotated sequences such as the Fluo-C2DL-Huh7, the temporal augmentation improves the segmentation quality. The improvement for the DIC-N2DL-HeLa is less notable.

H. Cell Segmentation Benchmark Results

The potential applicability of the method to numerous microscopy datasets is illustrated in the results for the CSB. Overall, 39 different methods have been submitted to the benchmark to date. Visualizations of the segmentation results are presented in Fig. 5. Note that despite the variability in cell appearance and density we trained

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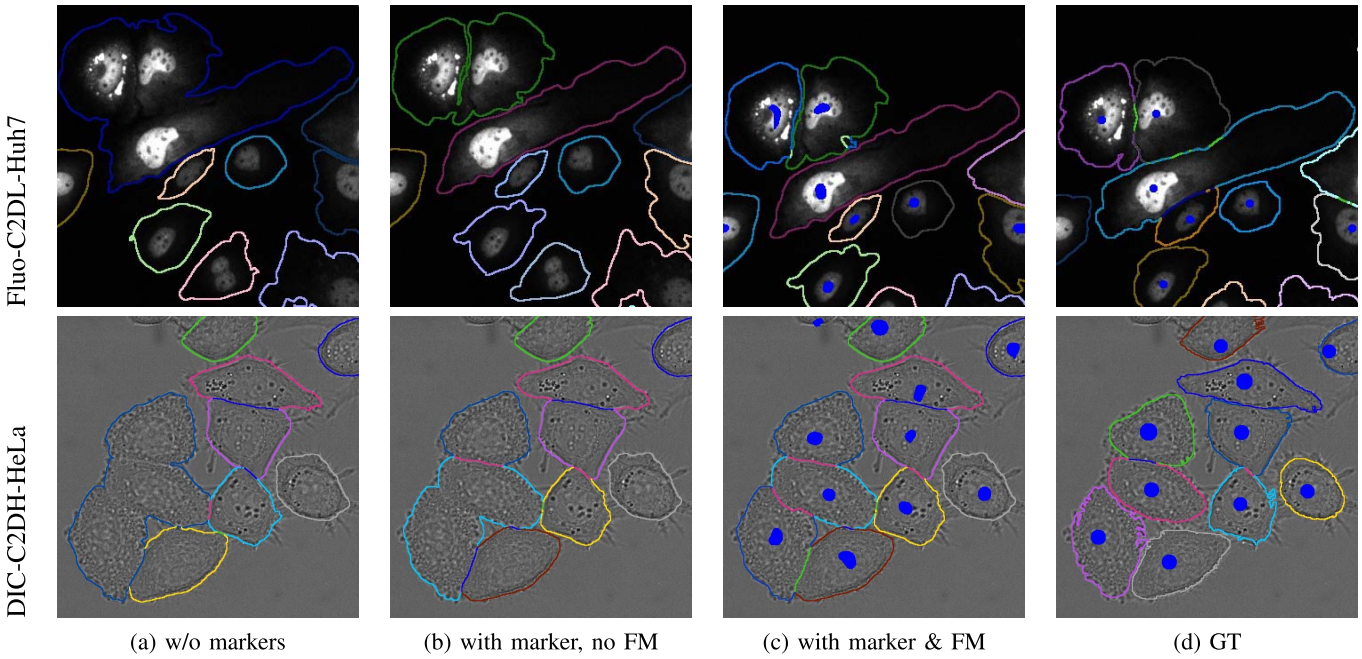

Fig. 4. Examples of the three experimental settings for the marker estimation experiment for a Fluo-C2DL-Huh7 and DIC-C2DH-HeLa **per-sequence test** image. The figure presents cell segmentation obtained when the network is trained without (a) and with (b) the marker estimation. (c) Image segmentation as in (b) where the markers are further utilized (via fast marching) to facilitate the separation of neighboring cell instances, the predicted markers are visualized in the image in blue. (d) Gold-truth cell outlines including the gold-truth markers, which are marked in blue.

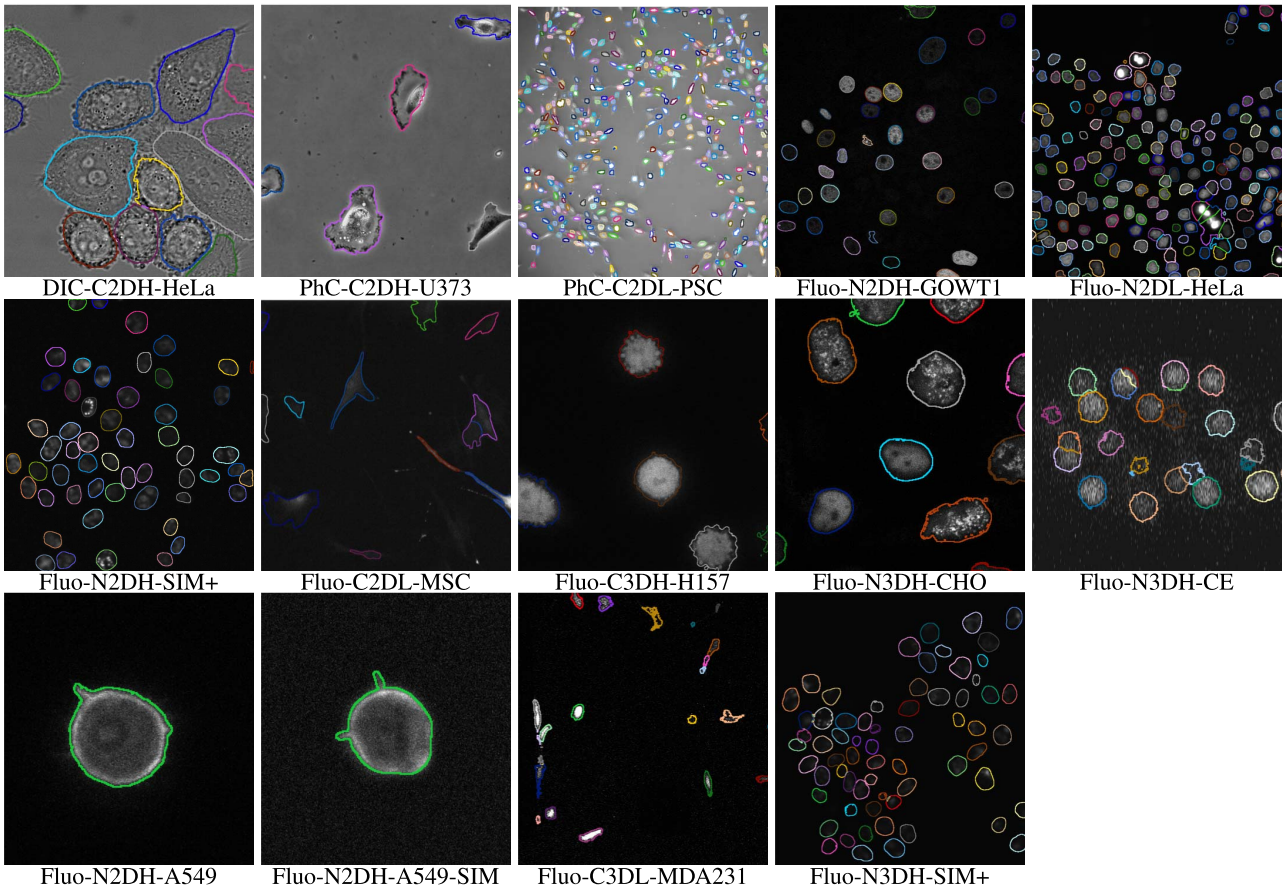

Fig. 5. Examples of the segmentation results for the submitted datasets. Each instance is marked by the segmentation contour. The color of the contour indicates the label. The 3D datasets are presented by the center slice.

the proposed network with the same hyperparameters for all datasets. The quantitative results and algorithms for all top-3 leading methods are publicly available on the CSB web site at `http://www.celltrackingchallenge.net/latest-csb-results/`. The table shown in the website presents the segmentation results for the 30 competing

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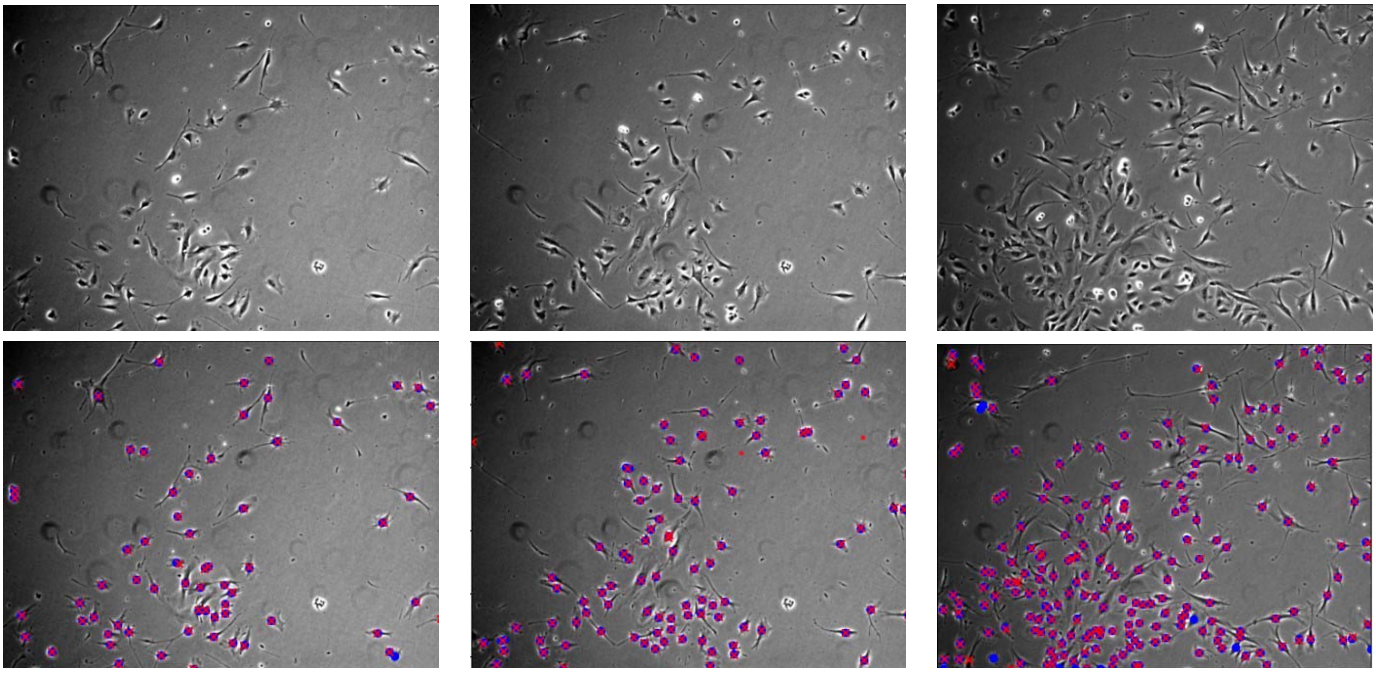

frame 300 frame 400 frame 680

Fig. 6. Examples of three test images exemplifying the appearance and the density of the C2C12 cells along with the detection results. The first row presents the raw data the second row depicts also the GT annotations (red crosses) and the predicted cell markers (blue circles).

TABLE V

OUR SCORES FOR SIX CSB DATASETS FOR WHICH OUR METHOD WAS RANKED IN THE TOP-3 (UPDATED TO NOV. 2021). OUR RESULTS ARE MARKED IN BOLD AND THE SCORE DIFFERENCE TO THE BEST PERFORMING METHOD (WHEN RELEVANT) IS INDICATED IN PARENTHESIS. THE SCORES OBTAINED BY OUR METHOD FOR ALL SUBMITTED DATASETS AND OUR MOST UPDATED RANKS APPEAR IN THE CSB WEBPAGE AT HTTP://CELLTRACKINGCHALLENGE.NET/PARTICIPANTS/BGU-IL/

DataSet	Measure	Rank	1_{st}	2_{nd}	3_{rd}
Fluo-C2DL-Huh7	OP_{CSB}	**2**/8	0.879 **(0.025)**	**0.854**	0.843
	SEG	**2**/8	0.791 **(0.040)**	0.752	**0.751**
	DET	**2**/8	0.968 **(0.012)**	**0.956**	0.935
Flou-N2DL-HeLa	OP_{CSB}	**1**/41	**0.957**	0.954	0.953
	SEG	**1**/41	**0.923**	0.923	0.919
	DET	5/41	0.994 **(0.003)**	0.992	0.992
PhC-C2DH-U373	OP_{CSB}	**2**/31	0.959 **(0.003)**	**0.956**	0.955
	SEG	7/31	0.927 **(0.005)**	0.927	0.925
	DET	**2**/31	0.991 **(0.001)**	**0.990**	0.985
Fluo-C3DH-A549-SIM	OP_{CSB}	**2**/10	0.977 **(0.034)**	**0.943**	0.929
	SEG	**2**/10	0.955 **(0.069)**	**0.886**	0.859
	DET	**1**/10	**1.000**	1.000	1.000
Flou-N3DH-SIM+	OP_{CSB}	**2**/13	0.949 **(0.064)**	**0.885**	0.866
	SEG	**2**/13	0.906 **(0.12)**	**0.786**	0.759
	DET	**2**/13	0.992 **(0.008)**	**0.984**	0.974
Fluo-N2DH-SIM+	OP_{CSB}	6/38	0.905 **(0.018)**	0.897	0.897
	SEG	12/38	0.832 **(0.042)**	0.825	0.822
	DET	**1**/38	**0.983**	0.981	0.979

teams (as of October 2021) that were ranked in the top-3. Our method, presented in orange, is labeled BGU-IL (5). Table V presents our CSB scores as appeared on the benchmark organizers' website in November 2021 for six datasets for which our method was ranked in the top-3. Our results are marked in bold and the score difference to the best performing method (when relevant) is indicated in parentheses. The scores obtained by our method for all submitted datasets (17) and our most updated ranks appear on the CSB webpage at `http://celltrackingchallenge.net/participants/BGU-IL/`.

Readers are also referred to the inter-annotator variability results at `http://celltrackingchallenge.net/latest-csb-results/` which presents the variability of different raters for the segmentation of the real datasets. The GT segmentations for the non-simulated (real) datasets were generated by averaging the segmentations of the different annotators. Note that for some of the datasets the differences

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TABLE VI
C2C12 CELL DETECTION RESULTS COMPARISON OF CELL DETECTION PERFORMANCE ON C2C12 DATASET USING F1-SCORE, PRECISION AND RECALL, WITH PREVIOUS STATE-OF-THE-ART ALGORITHMS REPORTED FOR THIS DATASET

Method	F1-score	Precision	Recall
CMF [31]	0.934	0.968	0.902
MPM [32]	0.948	0.964	0.932
Hayashida [33]	0.956	0.972	0.940
Ours	**0.971**	**0.980**	**0.960**

between the human annotators are at about the same magnitude as the scores for comparison of the automatic results with the GT.

I. C2C12 Cell Detection Results

Next, we examined the accuracy of the proposed marker estimator, in the absence of GT cell segmentations. For this purpose, we applied the proposed dual-task CLSTM-UNet to the C2C12 dataset [41] where only the GT cell coordinates were available for training. We compared our cell detection results to previous state of the art algorithms reported for this dataset: [31], [32] and [33].

1) Training: For the training we used 400 annotated frames from four different sequences (100 frames per sequence from Control, BMP2 and two sequences from FGF2 + BMP2) as described in [32]. Since the annotations are point level we used a Gaussian filter with kernel size of 51 and $\sigma = 6$ in order to dilate the cell centers. We then applied a threshold to receive a binary map of cell markers (GT marker map) that was used to define the binary cross entropy loss for the marker output of the network. The categorical cross entropy loss for the cell segmentation output of the network was calculated with respect to a trinary representation of the generated GT marker map, where each marker was further expanded. Note that the actual cell shapes (as shown in Fig. 6) is different than the expanded circular markers.

2) Test: With no GT segmentation available our detection results were exclusively based on the marker estimator. As in [31]–[33] we evaluated the results using a fully annotated sequence of the C2C12 dataset used for control. Table VI presents the precision, recall and F1 score measures as reported in [33] in comparison to these measures calculated based on our results. It should be noted that we have not performed any adaptation to the network structure or the training regime to fit the unique C2C12 data and the detection problem. We also used the same hyperparameters and training data augmentation as we used for the CSB data and did not apply any post-processing. Nevertheless, it can be clearly seen from the table that our method outperforms all others.

VI. SUMMARY

We presented a unified framework for the automatic instance segmentation of time-lapse microscopy sequences, in both 2D and 3D, which can utilize cell dynamics as well as sparse and weak annotations. One of the key ideas was triggered by our observation that human annotators often rely on temporal cues to accurately separate neighboring cells and detect partially visible cells. Our framework is based on a unique network architecture that incorporates ConvLSTMs into the well-established U-Net network, allowing it to extract meaningful spatio-*temporal* features at multiple scales and propagate them through time. The proposed ConvLSTM-UNet is structured and trained as a dual-task network having a complementary marker loss that facilitates the instance segmentation task. The incorporation of approximate cell markers, as opposed to full segmentations alone, not only improved the final segmentation but also relaxed the segmentation requirements, thus making data annotation a feasible task. We further showed the significant impact of the 3D input configuration. This configuration uses a fraction of the computational load required by a full 3D convolutional network and also allows the annotator to segment individual slices and not the full 3D segmentation. Finally, we exploited the built-in marker estimator for the task of cell detection in the complete absence of GT cell segmentation.

The ConvLSTM-UNet efficiently utilizes sparse, weak, and temporal annotations that enable the network to accurately detect and segment cells in difficult scenarios. Quantitative analysis showed that our method achieves competitive results (Table V), ranking in the top-3 for six different 2D and 3D datasets in the Cell Segmentation Benchmark. We also presented in Table VI state-of-the-art cell detection results for an additional public dataset with weak annotations.

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